CARDIFF UNIVERSITY
PRIFYSGOL CAERDYDD

Do not turn this page over until instructed to do so by the Invigilation Supervisor

Academic Year: 2024/25

Examination Period: Spring

Module Code: MAT700

Examination Paper Title: Mathematical Methods for Data Mining

Duration: 2 hours

Please read the following information carefully:

Structure of Examination Paper:

- There are 4 pages including this page.
- There are **4** questions in total.
- There are no appendices.
- The maximum mark for the examination paper is 75 and the mark obtainable for a question or part of a question is shown in brackets alongside the question.

Instructions for completing the examination:

- Complete the front cover of any answer books used.
- This examination paper must be submitted to an Invigilator at the end of the examination.
- Answer **THREE** questions.
- Each question should be answered on a separate page.

You will be provided with / or allowed:

- **ONE** answer book.
- Squared graph paper.
- The **use of calculators** is **permitted** in this examination.
- The use of a translation dictionary between English or Welsh and another language, provided that it bears an appropriate departmental stamp, is permitted in this examination.
- The use of the student's own notes, up to **1 sheet (2 sides) of A4 paper**, is permitted in this examination.

1. (a) i. Explain the Bonferroni's principle. [4]

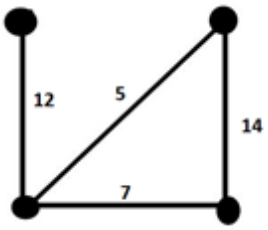
ii. Let us look at a class of 30 students and let us assume that their birthdays are independent and uniformly distributed over the 365 days of the year. We are interested in the event that some students have their birthday on the same day. Show that the expectation of the number of pairs of students having the same birthday in a class of 30 is ≈ 1.192. [4]

(b) i. Define TF×IDF score for words in a collection of documents. [4]

ii. Suppose there is a repository of 256 documents, and a word w appears in 64 of them. In a particular document D, the maximum number of occurrences of a word is 45. What is the TF×IDF score for w in D if that word appears six times in D? [4]

iii. What is the document-term matrix for a corpus of documents? [3]

(c) i. Find the adjacency matrix of the following weighted graph: [2]



ii. Let $G = (V, E)$ be an undirected graph. Show that [4]

$$\sum_{v \in V} \deg(v) = 2|E|.$$

2. (a) Prove that the L_∞ distance is indeed a distance measure. [4]

(b) Prove that the *edit distance* is indeed a distance measure. [5]

(c) Define the Jaccard distance between sets. Prove that this function is a distance measure. [6]

(d) Describe the Naïve Bayesian classification. [5]

(e) Describe the most common mathematical framework for learning. [5]

3. (a) Consider the one-dimensional data set x with labels y shown in the Table

x	1	4	9	16	25	36	49	64	81
y	+	+	-	-	-	**?**	+	-	+

i. Compute the class of x = 36 according to its 2-, 3-nearest neighbours (using the majority vote). [5]

ii. Repeat the previous task, but use the weighted version of kNN. [5]

(b) Define the *softmax* function and the *cross entropy* between two probability distributions over the same underlying set of events. [5]

(c) Describe the k-means clustering algorithm and its objective function. How to select initial centroids for the k-means algorithm? [5]

(d) What are spider traps? Explain how the PageRank

$$\nu = \big((1-\beta)\frac{E}{n} + \beta M\big)\nu$$

model overcomes the problem of spider traps. [5]

4. (a) Describe how to calculate *precision*, *recall* and *F-measure* of a binary classifier. Prove that F-measure of any binary classifier is $\leq \frac{\text{precision}+\text{recall}}{2}$. [5]

(b) What is a meta-algorithm? Describe the halving algorithm and the weighted majority algorithm. [5]

(c) Suppose we have access to a set of n experts and assume that one of the experts is consistent. Prove that the halving algorithm makes at most $\log_2 n$ mistakes. [5]

(d) Compute the PageRank of each page in Fig. 1, assuming $\beta = 0.8$. [10]

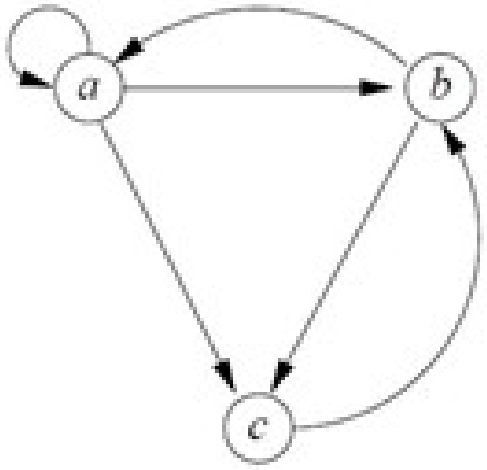


Fig. 1